

Lab-6 Ex2 - Factorial Again! (Prelab*)

This is a PreLab exercise. Please do the PreLabs after each lecture and submit before you come to lab!

This is the same problem we have done last time! This time, let's try to use ***for-loop***.

In this exercise, let's try to write a C program that calculates a factorial. For a positive integer x , the factorial of x , expressed in $x!$, is given by:

$$x! = (x) * (x-1) * \dots * 1$$

To use looping to solve the problem, we can set x to be our loop variable, and decrease it by 1 every time until x reaches the value of 1.

Assume the input is in range of 1 to 15. User-input integers are ***bold and italic*** in the following sample runs.

Sample run #1:

5

120

Sample run #2:

15

2004310016

Sample run #3:

1

1

*P.S. Do you know why we restrict the range of input x ? Use a calculator to get the answer for ***15!***. Do you think we should restrict the range further?*

(NOTE: Please follow the pseudocode to complete the program anyhow)